Combinatorics — Complete Formula Reference

Every identity from your syllabus, zero derivations — open this during the exam, not the worked-problem books.

1. Fundamental Counting

- **Sum rule:** $|A \cup B| = |A| + |B|$ if $A \cap B = \varnothing$
- **Product rule:** independent sequential choices multiply
- **Permutations:** $P(n,k) = \dfrac{n!}{(n-k)!}$
- **Combinations:** $\binom{n}{k} = \dfrac{n!}{k!(n-k)!}$
- **Multinomial:** $\binom{n}{n_1, \ldots, n_r} = \dfrac{n!}{n_1! \cdots n_r!}$, $\sum n_i = n$

2. Binomial Identities

- **Pascal:** $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$
- **Symmetry:** $\binom{n}{k} = \binom{n}{n-k}$
- **Binomial thm:** $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$
- **Vandermonde:** $\sum_{j=0}^{r} \binom{m}{j}\binom{n}{r-j} = \binom{m+n}{r}$
- **Hockey-stick:** $\sum_{i=r}^{n} \binom{i}{r} = \binom{n+1}{r+1}$
- **Row sum:** $\sum_{k=0}^{n} \binom{n}{k} = 2^n$

3. Circular & Restricted Selections

- **Circular perm. (rotations $\equiv$):** $(n-1)!$
- **Circular perm. (rot. & reflection $\equiv$):** $\dfrac{(n-1)!}{2}$, $n \geq 3$
- **Non-adjacent, in a row** (choose k of n): $\binom{n-k+1}{k}$
- **Non-adjacent, on a circle:** $\dfrac{n}{n-k}\binom{n-k}{k}$

trap: on a circle, never use the row formula — the $\frac{n}{n-k}$ factor is not optional.

4. Stars and Bars

- **Non-neg. solutions** $\sum x_i = n$, $x_i \geq 0$, k vars: $\binom{n+k-1}{k-1}$
- **Positive solutions** $x_i \geq 1$ (composition of n into k positive parts): $\binom{n-1}{k-1}$
- **General lower bound** $x_i \geq a_i$: sub $y_i = x_i - a_i \geq 0$, apply above to target $n - \sum a_i$
- **Capacity cap** $x_i \leq c$: (unrestricted count) − PIE-corrected overflow at $x_i \geq c{+}1$ per variable

trap: $\binom{n+k-1}{k-1}$ ($x_i \geq 0$) vs. $\binom{n-1}{k-1}$ ($x_i \geq 1$) is the #1 mix-up — know which one your problem needs before you compute.

5. Inclusion–Exclusion

- **Two sets:** $|A \cup B| = |A| + |B| - |A \cap B|$
- **Three sets:** $|A \cup B \cup C| = \sum |A_i| - \sum |A_i \cap A_j| + |A \cap B \cap C|$
- **General (N sets):** $\left|\bigcup A_i\right| = \sum_{\varnothing \neq S} (-1)^{|S|+1} \left|\bigcap_{i \in S} A_i\right|$
- **Boole's inequality:** $P(\bigcup A_i) \leq \sum P(A_i)$
- **Bonferroni:** alternating partial PIE sums sandwich the true value (over-, under-, over-, …)

trap: "divisible by a and b" means $\operatorname{lcm}(a,b)$, never $a \times b$, unless $\gcd(a,b) = 1$.

6. Derangements

- **Formula:** $D_n = n! \sum_{i=0}^{n} \dfrac{(-1)^i}{i!}$
- **Recurrences:** $D_n = (n-1)(D_{n-1} + D_{n-2}) = nD_{n-1} + (-1)^n$
- **Large-n:** $D_n \approx n!/e$; $P(\text{no fixed pts}) \to 1/e \approx 0.3679$
- **Exactly k fixed points:** $\binom{n}{k} D_{n-k}$
- **Table:** $D_0{=}1, D_1{=}0, D_2{=}1, D_3{=}2, D_4{=}9, D_5{=}44, D_6{=}265, D_7{=}1854$

trap: "no fixed points" ≠ "no fixed points and no 2-cycles." The latter needs the cycle-type formula in Sec. 8, not D_n.

7. Catalan Numbers & Ballot Theory

- **Closed form:** $C_n = \dfrac{1}{n+1}\binom{2n}{n} = \binom{2n}{n} - \binom{2n}{n+1}$
- **Recurrence:** $C_{n+1} = \sum_{i=0}^{n} C_i C_{n-i}$, $C_0 = 1$
- **Generalized ballot** (p ups, q downs, $p \geq q$, stay ≥ 0): $\dfrac{p-q+1}{p+1}\binom{p+q}{q}$
- **First return to 0 at step** $2k$ (semilength n): $C_{k-1}\, C_{n-k}$
- **Values (n=0..10):** $1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796$

8. Compositions, Partitions & Cycle Types

- **Compositions, k positive parts:** $\binom{n-1}{k-1}$
- **Compositions, k non-neg. parts:** $\binom{n+k-1}{k-1}$
- **All compositions of n (any # parts):** 2^{n-1}
- **Partition function $p(n)$:** no closed form — enumerate for small n
- **Euler's theorem:** #(partitions, odd parts) = #(partitions, distinct parts) [Glaisher bijection: binary-expand each multiplicity]
- **Cycle-type count** (type $1^{a_1} 2^{a_2} \cdots$): $\dfrac{n!}{\prod_i i^{a_i} a_i!}$
- **n-cycles on n labeled elements:** $(n-1)!$

Quick Decision Guide — "Which Formula Do I Need?"

- Order matters, no repeats → **permutation** $P(n,k)$.
- Order doesn't matter, no repeats → **combination** $\binom{n}{k}$.
- Splitting into labeled groups of fixed sizes → **multinomial**.
- Identical items into distinguishable bins, order irrelevant within a bin → **stars and bars** (check ≥ 0 vs. ≥ 1 vs. capped).
- "at least one of / none of" several overlapping conditions → **PIE**.
- "no element maps to itself" → **derangements**; if *also* no 2-cycles (or longer forbidden cycles) → **cycle-type formula**, not D_n.
- Balanced parentheses / non-negative running balance / non-crossing matchings → **Catalan number** (equal steps) or the **ballot formula** (unequal steps).
- "In how many *orders*" with positive parts summing to n → **composition**. "In how many *ways*, order irrelevant" → **partition** (usually hand-enumerate).
- Two differently-described sets counted are claimed equal → look for a named **bijection theorem** (Euler, Glaisher, Catalan-family) before brute-force counting either side.

Numbers Worth Memorizing Cold

- $\binom{2n}{n}$ for $n = 1..6$: $2, 6, 20, 70, 252, 924$
- C_n for $n = 0..8$: $1, 1, 2, 5, 14, 42, 132, 429, 1430$
- D_n for $n = 0..7$: $1, 0, 1, 2, 9, 44, 265, 1854$
- $n!/e$ rounded, $n = 6, 7, 8$: 265, 1854, 14833 (matches D_n exactly)
- $1/e \approx 0.3679$; $e \approx 2.71828$